

Google's New Android App Bundle: A Win-Win for Developers and Users



Android App Bundle is a new app publishing format (*.aab*), which enables Play Store to publish lean app APKs (Android Package Kits). So developers now need to code less when pushing an app to Play Store, and users benefit by getting a small-sized APK, which helps them save space on the phone. Let's take a quick look at this new format.

One of the most stressful tasks of an Android developer is managing the distribution of an app, uniformly, on all devices across the globe. There are concerns about resource management, screen density, CPU architecture, language management, etc. In the current scenario, multiple APK files (comprising all types of

resources for the app's target devices), which satisfy all of the app's requirements, must be developed for it to run flawlessly and efficiently. The price for this is a huge APK size. Figure 1 represents the APK format system.

Google has addressed this issue by introducing the Android App Bundle format. This new upload format is very similar in code to the APK, but differs when you sign in and upload it to Google Play.

The magic of App Bundle happens when a user approaches Play Store to download the application. Google Play analyses the requirements for the user's phone and only provides the resources needed for the smooth execution of the app on that device. The developer can now relax since there is no longer any need to build multiple APKs to support all devices, while the user benefits by getting a small-sized APK, thereby saving phone storage space and data usage. This is a win-win situation for everyone.

Figure 2 represents the Android App Bundle format. Here the APK is generated by Google Play, with only the required resources for the target device. In this example, language support for English (EN), resources of xhdpi and 64-bit ARM CPU architecture are the only contents of the APK file.

Also, Google allows the developer to add dynamic features and modules to the application using the App Bundle. This means that users can download different modules of



Figure 1: APK file with the resources for all devices